

Alessandro Puddu

Senior Unity Engineer

Games, VR & Real-Time 3D

PROFILE

Senior Unity Engineer with 5+ years delivering **commercial games**, **enterprise real-time 3D applications**, and **confidential client products**. A pragmatic generalist owning architecture and delivery across **PC, WebGL, VR/XR, and mobile**. Combines broad Unity experience with hands-on **LLM and speech integrations** and extensive use of **coding agents** within reviewed and tested development workflows. Available for long-term international engagements, embedded senior roles, project rescue, and end-to-end delivery.

SKILLS

- **Unity & Platforms:** Unity, C#, PC, WebGL, mobile, OpenXR, XR Interaction Toolkit, Addressables, cross-platform porting, profiling, optimization.
- **Gameplay & Simulation:** Gameplay systems, NPC behavior and game AI, state machines, physics and simulation, input, UI, animation, IK, rendering and shaders.
- **Architecture & Tooling:** Async workflows, modular state management, platform strategies, persistence, custom editor tools, diagnostics, testing, production debugging.
- **Networking & Integrations:** Unity Netcode, Mirror, REST APIs, WebSockets, SSE, authentication, streamed audio, real-time service integration.
- **Full Stack & Workflow:** TypeScript, Node/Express, Next.js, Prisma, PostgreSQL, technical documentation, code review.
- **AI & Speech Integrations:** LLM integration, conversational state, TTS/STT, streamed audio, browser/native speech interfaces.
- **AI-Assisted Engineering:** Coding agents, context engineering, task decomposition, repository instructions, AI-generated code review, automated validation.

EXPERIENCE

- **Independent Senior Unity Engineer** Remote / International
Solid Code Interactive December 2025 – Present
 - **Squirrel Games — Unity Lead (Contract)** March 2026 – Present
 - Lead technical direction and hands-on development of a cross-platform Unity application, spanning architecture, gameplay, XR, networking, tooling and service integrations.
 - Integrate LLM-backed conversational systems with Unity, including streamed responses, conversation state, runtime context and coordination with interactive character behaviour.
 - Develop text-to-speech (TTS) and speech-to-text (STT) integrations, including streamed audio playback, microphone input, provider abstractions and platform-specific browser and native implementations.
 - Implement client-backend integrations and authentication flows, handling asynchronous communication, cancellation, session lifecycle and error recovery across desktop, WebGL and VR.
 - Build character interaction and animation systems, including inverse kinematics, physics-driven reactions, camera behaviour and editor tools for content authoring.
 - Develop Meta Quest VR/MR functionality using OpenXR and XR Interaction Toolkit, including passthrough, hand tracking, controller/hand input switching, locomotion and spatial UI.
 - Optimise standalone VR rendering and runtime performance through platform-specific graphics settings, shader changes and resource management.
 - Apply coding-agent workflows to implementation, debugging and refactoring, maintaining repository guidance and using code review and automated validation to assess generated changes.
 - **Sandbox VR — Unity Developer (Contract)** May 2026 – Present
 - Contribute to *RaceClub*, a Unity VR product, with primary responsibility for UI Toolkit integration across runtime menus and HUD systems, plus gameplay feature delivery. Implement cross-device input and feedback flows, VR state fixes, and data-driven gameplay/UI behavior.

- Capgemini Engineering** Bologna – Remote
Unity Software Engineer (R&D) October 2023 – February 2026
 - Diagnosed and refactored CPU-heavy physics and simulation systems to achieve stable real-time performance on constrained target hardware.
 - Engineered Addressables-based content delivery to decouple assets from application releases and support secure on-demand loading.
 - Delivered confidential WebGL and VR applications with in-app authoring tools, real-time multiplayer, and platform-specific interaction models.
 - Ported content and runtime systems across WebGL, mobile, and VR constraints; built node-based editor tooling and optimized persistence for browser limits.
- Camera Interactive** Remote
Game Programmer March 2021 – September 2023
 - Contributed to the commercial Action-RPG *Alaloth: Champions of the Four Kingdoms*, shipping production gameplay and UI features for PC.
 - Developed in-engine tools for asset validation and world management, replacing repetitive manual production operations.
 - Implemented gameplay loops, inventory and skill systems, narrative events, UI architecture, and production fixes in close collaboration with design and QA.

SELECTED PUBLIC PROJECTS

- Alaloth: Champions of the Four Kingdoms** — Steam / PC, Game Programmer, 2021–2023. [Steam]
 - Shipped gameplay systems, UI architecture, narrative events, inventory and skill features in a commercial Action-RPG codebase.
 - Built asset-validation and world-management tools and supported production issue resolution with design and QA.
- Apocalypse Rangers** — PC / Online, Gameplay & Network Programmer, 2020–2021. [Trailer]
 - Built Mirror-based client-server multiplayer and online gameplay systems.
 - Implemented custom HLSL toon and terrain rendering; New Game Designer 2020, 3rd Place Online.
- Hell for One** — PC / RTS Hybrid, AI & Gameplay Programmer, 2019–2020. [Itch.io]
 - Designed NPC decision logic with state machines, steering behaviors, and reusable gameplay actions.
 - Implemented scalable battle spawning and coordination; New Game Designer 2020, 1st Place Offline.
- RuNeglect** — Microsoft HoloLens, Sole Developer, 2018–2019.
 - Built a rehabilitative AR thesis application using spatial mapping, object pooling, and room-adaptive gameplay.

ENGAGEMENT STRENGTHS

- **Codebase ownership:** map unfamiliar systems quickly, identify delivery risks, add diagnostics, and leave decisions understandable to the team.
- **Embedded collaboration:** work directly with product, design, art, and QA while remaining hands-on with architecture and implementation.
- **Cross-platform delivery:** separate platform-specific input, UI, rendering, persistence, and performance concerns across WebGL, VR, desktop, and mobile.
- **End-to-end problem solving:** move from Unity runtime work into editor tools, networking, APIs, authentication, or data systems when the product requires it.

EDUCATION

- **Università degli Studi di Milano**, Master's Coursework in Computer Science 2019–2021
 Advanced graphics, real-time rendering, artificial intelligence, and game development.
- **Università degli Studi di Milano**, B.S. in Computer Science 2019
Thesis: Development of an AR application for rehabilitation in hemispatial neglect (Microsoft HoloLens).

LANGUAGES

- **Italian:** Native **English:** Professional working proficiency